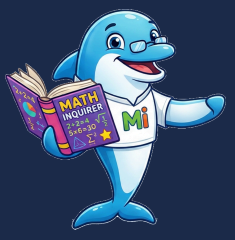
TUTOR GUIDE • LEVEL 1 • TERM 4 • LESSON 4 OF 10

Cubes and Views



Big idea: The same thing looks different from the top, from the front and when it is folded. A picture can hide a cube, but the numbers on a top view count every cube. A folded paper has layers, so one hole can make more holes when it is opened.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

1. At a glance

Strand	Not a listed Primary 1 outcome. 3-D shapes are Primary 2 (p. 34); drawing cubes on an isometric grid is Primary 5 (p. 42) (Counting cubes in a picture and in a top view (early number work; solid shapes are Primary 2, p. 34); Reading a top view as numbers and the cube you cannot see (beyond Primary 1; isometric drawing of cubes is Primary 5, p. 42); Fold once and punch once, then fold twice (not in the syllabus; the nearest idea is symmetry, Primary 4)) Strand label for the tutor only. This is an added lesson (new topic) and it is not a listed Primary 1 content outcome. It builds the visualising skill (stacks, views, folds) that is also used in contest-style items. Start here is approachable with blocks in hand: count the cubes in a picture, count the cubes you cannot see when the total is given, match a top view, write a top view from a picture, say where the fold is and read a top view as a total. BEYOND PRIMARY 1 ON PURPOSE at Think harder and Go further only: finding a missing number on a top view when the total is given, fold once and punch once (the holes are the same distance from the fold), the other hole on a grid, fold twice and punch once (each fold doubles the layers: 4 holes), and a stack with a hidden cube. Named puzzle: Fold Twice, Punch Once. Owner decisions, 22 and 23 Oct 2026: strong content with scaffolding, said plainly to families; every lesson names a beyond item; fold once and punch is Think harder (decision 6). Do not describe the lesson as MOE-aligned. Wording rule (Term 1 audit, pre-flight D18): wrong working is always 'a child' or 'a child in another class', never a cast name. There is no school-paper evidence for stacks or folds, so the wording is unverified. Non-routine types (tutor only): fold once and punch once, a view from the top, fold twice and punch once.
Objectives	• Count the cubes in a picture and say how many cannot be seen when the total is known. • Write a top view as a grid of numbers from a stack, and read a top view as a total. • Fold a paper square once and say where the fold line is. • Think harder: find a missing number on a top view, say how many holes a folded paper has when it is opened, and find the other hole on a grid. • Go further: fold twice and punch once, count a bigger stack with a hidden cube, and build a stack from a total.
Joining cold	**Remember warm-up covers:** Counting by rows (Term 2 Lesson 5, taught again with a picture); Doubles (Term 2 Lesson 3, taught again with a picture); Counting cubes in a picture, one touch each (taught with a finger). Only counting to 12 and adding two small numbers. Nothing from the lessons before is needed: the warm-up re-teaches counting by rows, doubles and counting cubes in a picture. The words stack, top view, fold and layer are taught in Explore and the Mini-lesson. Fairly safe to miss: no other lesson needs it, but Mi's Challenge Quest in the last lesson of the year has a gate on stacks and folds. A child who misses it watches the recording and does workbook pages 1 and 2. A child who joins here cold can follow: the warm-up re-teaches rows, doubles and counting cubes, and Start here keeps to stacks of at most 5 cubes.

Child materials	• Your Term 4 book (Book 7), workbook pages 1 to 5, and the fold-squares page (page 6): the 4 paper squares are cut out by a grown-up beforehand • 12 small blocks of one size: building bricks or dice. No blocks? Use buttons or coins on a 2 by 2 grid drawn on paper: the number of buttons in a square is the height of that stack • No printer or no cutting? Any square of paper works, or a sheet of paper as it is • A pencil, a rubber and spare paper
Tutor materials	• The slide deck • Your own 12 blocks and a paper square to show on camera, if you wish • A visible clock Nothing needs to be bought. Blocks and dice are small pieces: they stay on the table; nothing goes in the mouth or ears; keep them away from younger brothers and sisters. No matches, toothpicks or anything sharp. Nobody cuts during the lesson: a grown-up cut the squares beforehand. Nobody punches a real hole: the hole is a pencil dot made on the folded paper.
Tiers	Start here (6 questions, pages 1 and 2), Think harder (3, page 3), Go further (3, page 4), Investigation and look-back (page 5), kit page (page 6). Labels are neutral; nobody is told which page they are on.
Beyond Primary 1 (tutor only)	Fold Twice, Punch Once at G1, G2: A paper square is folded in half and then in half again, and one hole is punched through all the layers: how many holes when it is opened (4, because each fold doubles the layers), and which opened paper it makes. Also a stack on a 2 by 2 base with a cube hidden behind a taller tower (3, 1, 2 and 2 make 8). Neither is in the Primary 1 syllabus: 3-D shapes are Primary 2 (p. 34), drawing cubes is Primary 5 (p. 42) and folding for symmetry is Primary 4. Fold once and punch once is the Think harder step. Named puzzle: Fold Twice, Punch Once. Never in Start here, never advertised to children.
Contest-style move	Used at Think harder, Go further and the Investigation: fold once and punch once: how many holes, and where (the holes are the same distance from the fold); a view from the top: the numbers on a top view count every cube, even one you cannot see; fold twice and punch once: each fold doubles the layers (1, 2, 4); the hidden cube: the total on a top view minus the cubes you can see. Never advertised to children or parents.

Before the lesson

- Solve every workbook question yourself first. Check the answers: S1 5; S2 3 and 1; S3 B; S4 1, 3 and 1, 1; S5 a line across the middle; S6 7; T1 2 and 1; T2 2 and second; T3 a hole 1 step right of the line, and 1; G1 4 and third; G2 2 and 1; G3 any four numbers from 1 to 3 that make 7; I1 2 dots, the same distance from the fold.
- Tell parents a week before: the fold-squares page (page 6 of the book) needs 4 squares cut out by a grown-up, and the family finds 12 small blocks of one size (building bricks or dice); a family with no blocks uses buttons on a drawn grid, and the lesson works with pictures. The next lesson needs the pair-cards page of Book 7 torn out, with its 12 cards cut out.
- Have the slides open. Have your own blocks and a paper square to show on camera if you wish. Practise the arm-fold break. When you read a stack question, say 'the back stacks are in the top row' every time.
- Check the recording captures the tutor and the shared screen. Children's video is excluded; any unmuted voice is recorded. Greet the whole group first, say the notice, then start the recording; avoid first names while it runs.
- Be plain with families: counting cubes in a picture is early work; reading a top view as numbers, the cube you cannot see, fold once and punch once and fold twice go beyond Primary 1 on purpose, always with real blocks and paper. 3-D shapes are Primary 2 and drawing cubes is Primary 5. Do not call the lesson MOE-aligned.
- Wording rule: a picture cannot show every cube, so hidden-cube questions always give the total or the top view. Wrong working belongs to 'a child' or 'a child in another class', never to a cast name and never to a child in the group. Slides keep the answers off the screen while you ask: read every question with its numbers aloud.

Safeguarding and recording (this lesson)

- Blocks, dice and paper squares are small pieces. They stay on the table: nothing goes in the mouth or ears. No matches or toothpicks. Nobody cuts during the lesson; a grown-up cut the squares beforehand. Nobody punches a real hole: the hole is a pencil dot.
- This lesson uses the main room only: no breakout rooms, so the tutor sees every camera. A parent sits nearby but does not give answers.
- Nobody holds paper, blocks or a drawing up to the camera: a gallery view shows everyone. Children tick their own work against the answer slide, answer together on a count of three, or show fingers or a thumb for a choice. There is no private chat.
- Wrong working belongs to 'a child' or 'a child in another class', never to a child in the group, and never to a name. Never read out a child's name with a right or wrong answer. The recording has the tutor, the shared screen and any voice that is unmuted. Children may mute and keep their cameras off. A child's page is never shown.
- Movement break: children need a little clear space, soft shoes or socks on a safe floor, no stairs and no running. Moves are gentle; a child may stay seated and move one arm.
- A child who is still stuck ticks 'I am still stuck' on the look-back page. A grown-up may photograph that page after class and send it to the business WhatsApp only; photos are kept for 90 days after the program ends. The tutor then helps in the open session next lesson. No private messages to children, and no personal phones.
- The look-back is never collected in front of the group or read out. A pass is always allowed.

2. Timing plan (90 minutes exactly)

Minutes	Part	Min	Slides	Script fills
0–2	**Opening**	2	1: Cubes and Views	1:35
2–8	**Remember warm-up**	6	3: Warm-up: rows, Doubles, Count the cubes	5:00
8–12	**Sneak-peek puzzles**	4	1: Sneak-peek puzzles	3:00
12–23	**Explore**	11	5: Explore: build the stack, 6 blocks, 1 hidden, Explore: fold and open, Open it up, What did we notice?	9:52
23–34	**Mini-lesson**	11	10: Count every cube, 7 cubes in all, The top view as numbers, The top view, Build from the top view, 6 cubes, Fold once, punch once, Two holes, Same distance from the fold, 2 steps each side	10:10
34–38	**Brain break**	4	1: Fold and unfold break	3:16
38–70	**Workbook**	32	5: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Check page 2, Choose your next page	23:55
70–76	**Talk time**	6	2: Talk and explain, Back to the sneak-peek	4:26
76–82	**Exit ticket**	6	2: Exit ticket, Check the exit ticket	4:23
82–85	**Look-back**	3	1: Look-back	2:06
85–90	**Flex time and goodbye**	5	1: Well done, inquirers!	0:24
	Total	**90**		**68:06**

The script needs 68:06 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 80:43. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 12 (end of sneak-peek), minute 38 (start of workbook), minute 70 (talk time). If more than 2 minutes behind at minute 38, skip in this order: Start with: shorten Explore round 3 (what did we notice) to one question; Then: cut the other hole on a grid (s-m5); Then: take the extra minutes from the flex time. Never skip the look-back or the camera and privacy moments.

3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

0:00 – 2:00 Opening

script 1:35 of 2:00

0:00	SLIDE	Show slide **Cubes and Views** (s-title)
0:10	DO	Greet the whole group warmly, before the recording starts. Check that you can see each child, or ask the grown-up. [0:12]
0:22	SAY	Hello, inquirers! Pencil ready, and your blocks and one paper square on the table. Cameras can be on or off, as you like.
0:34	SAY	This lesson is recorded so a friend who is away can catch up. The recording has me, the shared screen and any voice that is not muted. You may keep your microphone and camera off.
0:51	DO	Start the recording now. Do not use children's first names on the recording. [0:05]
0:56	REC	*If you are watching the recording: hello! Get a pencil, paper, your workbook, your blocks and a paper square. Pause or rewind any time.*
1:08	SAY	Microphone off until I ask you to speak. To answer, say it with me on three, show fingers, or give a thumbs up. Nobody holds paper up to the camera.
1:23	SAY	Today the same thing looks different from different places. We count cubes we cannot see, we look from the top, and we fold paper.
	NOTE	Say the recording statement and the microphone routine in the same words every lesson, for anyone who joined late. Say the notice before you start the recording, and do not use children's first names once it is running. The routine and the four inquirer rules were introduced earlier in the year: do not re-teach them.
	NOTE	Plain statement for families when it comes up: counting cubes in a picture is early work; 3-D shapes are Primary 2 and drawing cubes is Primary 5, so a top view read as numbers, the cube you cannot see and fold and punch go beyond Primary 1 on purpose, always with real blocks and paper. There is no school-paper evidence for these topics, so the wording is unverified.
	NOTE	New here? Say in one minute: our three rule words (notice it, test it, explain why); a guess is fine and you can always pass; after Start here you may choose Think harder, Go further or the Investigation; nobody holds paper to the camera. A child without blocks uses buttons or counters on a drawn 2 by 2 grid: the number of buttons in a square is the height of that stack.

2:00 – 8:00 Remember warm-up

script 5:00 of 6:00

2:00	SLIDE	Show slide **Warm-up: rows** (s-rem1)
2:12	REC	*Recording: join in! Say your answers out loud before I say them. Pause if you need longer.*
2:20	SAY	Counters in rows. Count by rows.
2:24	ASK	How many rows are there? How many counters are in each row? How many in all? [wait about 35 s for answers]
3:06	CHILDREN	Your turn. Three rows of two. How many in all? Say it with me. [0:35]
3:42	SLIDE	Show slide **Doubles** (s-rem2)
3:54	SAY	A double is the same number twice. Double three is three and three again.
4:00	ASK	What is double three? What is double five? [wait about 35 s for answers]
4:40	CHILDREN	Your turn. What is double four? Show with your fingers. [0:30]
5:10	SLIDE	Show slide **Count the cubes** (s-rem3)

5:22	**SAY**	Cubes in a picture. Touch each cube once.
5:26	**ASK**	How many cubes are there? [wait about 35 s for answers]
6:03	**CHILDREN**	Your turn. The same shape with one more cube on the small tower. How many cubes? [0:35]
6:38	**CHILDREN**	Say each answer with me: two rows, three in each row, six; six; ten; three. Now the your-turn answers: six, eight, four. Then stretch your fingers. [0:22]
	IF STUCK	If a child is unsure how to count the cubes, touch each cube once and say a number. For rows, run a finger along each row. For doubles, hold up the same number of fingers on each hand.
	NOTE	Answers: 2 rows, 3 in each row, 6 in all; double 3 is 6, double 5 is 10; 3 cubes. Your turns: 6; double 4 is 8; 4 cubes. Children answer with a whisper, fingers or hands. Nobody writes. Joining cold: all you need is counting to 12 and the pictures.

8:00 – 12:00 Sneak-peek puzzles

script 3:00 of 4:00

8:00	**SLIDE**	Show slide **Sneak-peek puzzles** (s-sneak)
8:12	**REC**	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*
8:22	**SAY**	Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start, and I do not know is welcome. Nobody is marked.
8:34	**READ**	Puzzle one. How many cubes are in the picture? Write the number.
8:41	**CHILDREN**	Children write on their own paper. [0:35]
9:16	**READ**	Puzzle two. The stack has five cubes. The picture shows only four. Where is the fifth cube? I know because blank.
9:27	**CHILDREN**	Children write or draw their answer. [1:25]
10:52	**SAY**	Keep your paper. Nobody shows it. We will check our own thinking later, at Talk time.
	NOTE	Puzzle 1 answer: 4 cubes. Puzzle 2 answer: it is hidden behind the front tower (the cube at the back left, at the bottom). Look for good ideas to praise later (a child who said there must be one behind). Do not write scores or keep a list.

12:00 – 23:00 Explore

script 9:52 of 11:00

12:00	**SLIDE**	Show slide **Explore: build the stack** (s-explore1)
12:12	**REC**	*Recording: pause now. Build the stack with your blocks, then fold your square and make one dot. Press play when you have tried.*
12:24	**SAY**	Today we are builders. Look at the stack on the screen. Build it with your blocks. Count how many blocks you use.
12:34	**CHILDREN**	Children build the stack. Tutor scans the cameras. [3:20]
15:54	**ASK**	How many blocks did you use? How many blocks can you see in the picture? [wait about 40 s for answers]
16:42	**SLIDE**	Show slide **6 blocks, 1 hidden** (s-explore1-ans)
16:50	**SAY**	Six blocks in all. The picture shows only five. One block is hidden behind the front tower.
16:58	**SLIDE**	Show slide **Explore: fold and open** (s-explore2)
17:10	**SAY**	Round two. Take your paper square. Fold it in half so the edges meet, and press the fold.
17:20	**READ**	Make one pencil dot on the folded paper, away from the fold. How many dots will there be when you open it?
17:32	**CHILDREN**	Children fold, make a dot and guess. Wait before anyone opens. [1:30]
19:02	**ASK**	How many dots do you think you will see? [wait about 25 s for answers]
19:31	**CHILDREN**	Now open your paper and look. Tell your grown-up what you see. [0:50]
20:21	**SLIDE**	Show slide **Open it up** (s-explore2-ans)

20:29	**SAY**	Two dots, one each side of the fold, the same distance from it.
20:36	**SLIDE**	Show slide **What did we notice?** (s-explore3)
20:44	**ASK**	What was hidden in the picture of the stack? Where were the two dots after you opened the paper? [wait about 45 s for answers]
21:38	**SAY**	A hidden block is still there, and the top view numbers count it. One dot on a folded paper makes two dots. Next we say it exactly.
	IF STUCK	A child who cannot build from the picture: build the front stack first, then the back. A child who puts the dot on the fold: make a new dot, away from the fold. A child with no blocks puts buttons on a drawn 2 by 2 grid. A child with no square folds any sheet of paper.
	NOTE	Main room only; you cannot see small blocks, so ask children to say how many they used and to show fingers. Answers: 6 blocks, 1 hidden; the dot makes 2 dots, one each side of the fold. Do not say the rule before the children do.

23:00 – 34:00 Mini-lesson

script 10:10 of 11:00

23:00	**SLIDE**	Show slide **Count every cube** (s-m1)
23:08	**REC**	*Recording: keep watching and copy the examples with your blocks and paper. Rewind if you miss a step.*
23:17	**SAY**	Look at this stack. Some cubes may be hidden.
23:22	**ASK**	How many cubes can you see in the picture? How many cubes are there in all? [wait about 60 s for answers]
24:30	**SLIDE**	Show slide **7 cubes in all** (s-m1-ans)
24:38	**SAY**	We can see six cubes. There are seven in all. The top view numbers add up: two, two, one and two. A cube we cannot see is still a cube.
24:52	**SLIDE**	Show slide **The top view as numbers** (s-m2)
25:00	**SAY**	A top view. Look from the top. Draw four squares like a window. Write how many cubes are in each stack. The back stacks go in the top row.
25:15	**CHILDREN**	Children write four numbers on their own paper. Wait 10 seconds before helping. [1:20]
26:35	**SLIDE**	Show slide **The top view** (s-m2-ans)
26:43	**SAY**	Top row one and one. Bottom row three and one. One, one, three and one make six cubes.
26:52	**SLIDE**	Show slide **Build from the top view** (s-m3)
27:00	**SAY**	Now the other way round. The top view tells us how many cubes are in each stack.
27:08	**ASK**	Build it with your blocks. How many cubes are there in all? [wait about 110 s for answers]
29:04	**SLIDE**	Show slide **6 cubes** (s-m3-ans)
29:12	**SAY**	Three, one, one and one make six cubes.
29:16	**SLIDE**	Show slide **Fold once, punch once** (s-m4)
29:24	**SAY**	A paper is folded in half. The fold is on the left. One hole is punched through both layers.
29:34	**ASK**	How many holes when it is opened? Where will they be? [wait about 50 s for answers]
30:30	**CHILDREN**	Children try it: fold the square, make one pencil dot, open it. [0:50]
31:20	**SLIDE**	Show slide **Two holes** (s-m4-ans)
31:28	**SAY**	Two holes. The fold is in the middle. The holes are the same distance from it, one each side.
31:37	**SLIDE**	Show slide **Same distance from the fold** (s-m5)
31:45	**SAY**	The dots stand for the opened paper. The dashed line is the fold. One hole is ringed.

31:54	**ASK**	How many steps is the ringed hole from the dashed line? Where is the other hole? [wait about 50 s for answers]
32:52	**SLIDE**	Show slide **2 steps each side** (s-m5-ans)
33:00	**SAY**	Two steps from the line, so the other hole is two steps on the other side, in the same row.
	IF STUCK	If a child counts only the cubes they can see, ask: what is behind? Build it with blocks. If a child writes the top view upside down, say the back stacks are in the top row. If a child puts the second hole on the same side, fold a paper and show that the holes come in pairs, one each side of the fold.
	NOTE	Key words to use every time: cube, stack, top view, fold, hole. Fixed question forms: How many cubes in all? How many holes when it is opened? The slides keep the answers off the screen while you ask: read every question with its numbers aloud. Class numbers are not used in the workbook.

34:00 – 38:00 Brain break

script 3:16 of 4:00

34:00	**SLIDE**	Show slide **Fold and unfold break** (s-break)
34:08	**REC**	*Recording: stand up and join in! Fold your arms in half, then open them.*
34:15	**SAY**	Stand up and get a sip of water. Fold your arms in half. Open them. Make a tall stack with your arms.
34:26	**CHILDREN**	Call movements: fold your arms in half, open them, stand tall, crouch low, make a square with your arms. Then bend in half again and again and say how many layers: two, then four. Seated children move one arm. [2:50]

38:00 – 70:00 Workbook

script 23:55 of 32:00

38:00	**SLIDE**	Show slide **Workbook page 1: Start here** (s-ws1)
38:08	**REC**	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*
38:17	**READ**	Page 1, Start here. Question 1: how many cubes are in the picture? Question 2: how many cubes are in the stack in all? How many cubes can you not see? Question 3: which top view matches the stack? Circle A, B or C.
38:41	**CHILDREN**	Page 1, questions 1 to 3. Tutor walks the cameras, asks how do you know. Wait 10 seconds before helping. [9:00]
47:41	**SLIDE**	Show slide **Check page 1** (s-ws1ans)
47:49	**READ**	Question 1: five cubes. Question 2: four cubes in all, and one you cannot see. Question 3: B. Tick your own page with a tick.
48:03	**SAY**	After question 3 you may choose: carry on, or try Think harder. Everyone may choose, at any time.
48:12	**SLIDE**	Show slide **Workbook page 2: Start here** (s-ws2)
48:20	**READ**	Page 2. Question 4: look at the stack. Write how many cubes are in each stack on the top view. Question 5: a paper square is folded so the top edge meets the bottom edge. Draw the fold line. Question 6: the top view shows how many cubes are in each stack. How many cubes are there in all?
48:52	**CHILDREN**	Page 2, questions 4 to 6. [8:20]
57:12	**SLIDE**	Show slide **Check page 2** (s-ws2ans)
57:20	**READ**	Question 4: one and three in the top row, one and one in the bottom row. Question 5: a line across the middle, from the left edge to the right edge. Question 6: seven cubes. Tick your own page.
57:41	**SLIDE**	Show slide **Choose your next page** (s-choice)
57:49	**SAY**	When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can swap at any time. Anything not finished is for home too.
58:05	**MOVE**	Stand and reach up for 30 seconds, then sit down and choose.

58:35	CHILDREN	Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5), or finish and check your own work. Tutor never says who is on which page. If Start here took the whole time, say that pages 3 to 5 are for home, and keep the time. [3:20]
	IF STUCK	Q1: touch each cube once. Q2: count the cubes you can see, then think what is behind. Q3: write the stack's numbers in a window and match. Q4: back stacks go in the top row. Q5: where do the top and bottom edges meet? Q6: add the four numbers.
	IF STUCK	A child who is still stuck ticks I am still stuck on the look-back page. The grown-up can photograph that page after class and send it to the business WhatsApp (never a personal phone). You then help in the open session in the next lesson. Never a private message to a child.
	NOTE	Think harder has a stack with a hidden cube, fold once and punch once, and the other hole. Go further has fold twice and punch once (beyond Primary 1 on purpose), a bigger stack with hidden cubes and building your own stack. Do not announce who is on which page. Start here takes a middle child about 17 minutes, so the choice time may be short. Slow class: take the extra minutes from the flex time and do not skip the answer slides.

70:00 – 76:00 Talk time

script 4:26 of 6:00

70:00	SLIDE	Show slide **Talk and explain** (s-talk)
70:08	REC	*Recording: pause. Tell a grown-up why counting only the cubes you can see can give too few.*
70:16	READ	A child in another class counts the cubes she can see in a picture. She sees six cubes. She says there are six cubes in the stack. Do you agree? How could we check?
70:35	CHILDREN	Think about it. You may use your blocks. [0:50]
71:25	CHILDREN	Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [0:40]
72:05	ASK	Who would like to share? Volunteers first. A pass is always fine. [wait about 100 s for answers]
73:51	SLIDE	Show slide **Back to the sneak-peek** (s-sneakans)
73:59	SAY	Remember the sneak-peek? Puzzle one: four cubes. Puzzle two: the fifth cube is hidden behind the front tower.
74:08	ASK	Did anyone change their mind while we worked? You can just nod. [wait about 12 s for answers]
	NOTE	Model answer: no, we do not agree. A picture can hide a cube behind another cube, so counting only what we see can give too few. We can check with blocks, or with a top view and add the numbers. Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed.

76:00 – 82:00 Exit ticket

script 4:23 of 6:00

76:00	SLIDE	Show slide **Exit ticket** (s-exit)
76:08	REC	*Recording: pause and do the exit ticket on your own. Then press play to check it.*
76:16	SAY	Exit ticket. Three questions, on your own, without looking back. Write your answers on page 5.
76:24	READ	E1. How many cubes are in the picture? E2. The top view shows how many cubes are in each stack. How many cubes are there in all? E3. Fold a paper square in half so that the left edge meets the right edge. Draw the fold line.
76:50	CHILDREN	Children answer E1 to E3 on their own. [3:10]
80:00	SLIDE	Show slide **Check the exit ticket** (s-exitans)
80:08	READ	E1: four cubes. E2: six cubes. E3: a line down the middle, from the top edge to the bottom edge.
80:19	SAY	Tick your own ticket with a tick. Well done.

	NOTE	Nothing is collected or marked live. Children tick their own ticket. If many children miss one question, spend two minutes on it at the start of the next lesson, in the open session, no names. Do not read out any child result.

82:00 – 85:00 Look-back

script 2:06 of 3:00

82:00	**SLIDE**	Show slide **Look-back** (s-look)
82:08	**REC**	*Recording: pause and fill in your look-back box. You can pass.*
82:14	**SAY**	Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you like, tick one thing to try next time. You can pass. Nobody reads it out. If you tick I am still stuck, ask your grown-up to take a photo of this page after class and send it to our business WhatsApp. Then we will help you.
82:46	**CHILDREN**	Children fill in the look-back box on page 5. [1:20]

85:00 – 90:00 Flex time and goodbye

script 0:24 of 5:00

85:00	**SLIDE**	Show slide **Well done, inquirers!** (s-end)
85:08	**SAY**	Well done, inquirers. Next lesson is Spot It. Please have the pair-cards page from your book, with the twelve cards cut out by a grown-up. Goodbye, and thank you to the grown-ups.
	SLIDE	Show slide **Go further: fold twice** (s-opt) *(optional)*
	SAY	Optional, only if you are on time. A paper is folded in half, then in half again. One hole is punched through all the layers. How many holes when it is opened?
	CHILDREN	Short think time, then take two answers. [1:30]
	SLIDE	Show slide **If we have time: pick one** (s-menu) *(optional)*
	SAY	Spare-time menu. If the whole class is ahead, pick one or two activities from the guide (about 4 minutes each).
	NOTE	This is the standing buffer. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: 4 holes, because each fold doubles the layers. If the whole class is ahead, use the Spare-time menu (guide section 3b) instead of rushing on.

3b. Spare-time menu: if the class is ahead

- Whole class ahead: use the Spare-time menu in the Flex time and the optional Go further slide. Pick one or two activities; do not start new content.
- A few children ahead: they may choose Go further or the Investigation, which are open to everyone. Never label or name a child, and never announce who is ahead.
- Never teach anything the next lesson needs. If the class is ahead, slow down: ask 'How do you know?' and let children explain, rather than rushing on.

Optional time, never added to the 90-minute sum. It comes from the 5-minute flex buffer, from slack in the parts, or from the optional Go further slide.

Activity	Minutes	You say	Recording cue	Answer
Is it always true?	4 (optional)	A child says: if every number on a top view gets 1 bigger, the stack has more cubes. Is that always true? Test it with a top view of 1, 2, 1 and 1, and then 2, 3, 2 and 2.	*Recording: pause. Test whether making every number on a top view 1 bigger gives more cubes.*	Yes. 1 + 2 + 1 + 1 = 5 and 2 + 3 + 2 + 2 = 9. Four numbers each 1 bigger add 4 more cubes.

Make a puzzle for your grown-up	4 (optional)	Build a stack with blocks. Tell your grown-up the number of cubes in all and show the picture or the top view with one number missing. Ask them to find it.	*Recording: pause. Build a stack and ask a grown-up to find the missing number on its top view.*	No single answer. Check by adding the numbers on the top view to get the total.
Teach it back	3 (optional)	Pretend a toy counts only the cubes it can see in a picture. Show the toy why a picture can hide a cube.	*Recording: pause. Show a grown-up how a cube can hide behind another cube.*	A cube behind a taller front tower cannot be seen from the front. The top view numbers or the blocks count it.

4. Questions, answers, hints and common mistakes

Sneak-peek puzzles (minutes 8 to 12)

Puzzle 1 (access): How many cubes are in the picture? Write the number.

Answer: 4 cubes.

Puzzle 2 (think): The stack has 5 cubes. The picture shows only 4. Where is the 5th cube? I know because ___.

Answer: The 5th cube is hidden behind the front tower.

Unmarked. Never put answers on boards in front of the group and never read a name with an answer. Reveal the answers where your script says (usually Talk time).

Start here (workbook pages 1 and 2)

S1 **Look at the stack of cubes.**

There are ☐ cubes in the picture.



Answer: 5 cubes

Hint (say this first): Touch each cube once and count.
Worked solution: Touch each cube once: 5 cubes. No cube is hidden in this picture.
Common mistakes: Counting a cube twice.; Counting the faces you can see instead of the cubes.
Time for a middle child: about 2.5 min.

S2 **The stack has 4 cubes in all.**

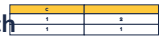
You can see ☐ cubes. You cannot see ☐ cube.



Answer: 3; 1

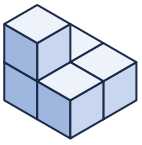
Hint (say this first): Count the cubes you can see. Take that number away from 4.
Worked solution: You can see 3 cubes in the picture. The stack has 4 cubes, so 1 cube is hidden. 3 and 1 make 4.
Common mistakes: Writing 4 for the cubes you can see.; Writing 0 because nothing looks hidden.
Time for a middle child: about 3 min.

S3 **Which top view matches this stack?**









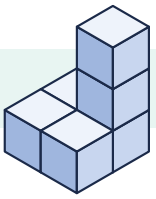
Circle A, B or C.

Answer: B

Hint (say this first): Back stacks go in the top row. How many cubes are in the back left stack?
Worked solution: The back row of the stack has 1 cube and 1 cube. The front row has 2 cubes and 1 cube. So the top row is 1, 1 and the bottom row is 2, 1. That is picture B.
Common mistakes: Choosing the view with the rows the wrong way up (A).; Putting the tall stack in the wrong place (C).
Time for a middle child: about 3 min.

S4 **Look at the stack. Write how many cubes are in each stack on the top view.**





Answer: 1 and 3 in the top row; 1 and 1 in the bottom row

Hint (say this first): The back stacks go in the top row. Count the cubes in each stack.
Worked solution: The back row has a stack of 1 and a stack of 3. The front row has a stack of 1 and a stack of 1. The top view is 1, 3 on the top row and 1, 1 on the bottom row.
Common mistakes: Writing the rows the wrong way up.; Writing the number of faces you can see.
Time for a middle child: about 3 min.

S5 **A paper square is folded so that the top edge meets the bottom edge.**
Draw the fold line on the square.



Answer: A line across the middle, from the left edge to the right edge

Hint (say this first): The fold is where the paper bends. Where do the top and bottom edges meet?
Worked solution: When the top edge meets the bottom edge, the paper bends halfway down. The fold line goes across the middle of the square, from the left edge to the right edge.
Common mistakes: Drawing the line down the middle.; Drawing a line across a corner.
Time for a middle child: about 2.5 min.

S6 **The top view shows how many cubes are in each stack.**
There are ☐ cubes in all.

2	3
1	1

Answer: 7 cubes

Hint (say this first): Add the four numbers. Start with the biggest.
Worked solution: 2 + 3 + 1 + 1 = 7. There are 7 cubes in the stack.
Common mistakes: Adding only two of the numbers.; Counting the squares (4) instead of the cubes.
Time for a middle child: about 3 min.

Think harder (page 3): the taught idea in a new setting

T1 **The stack has 6 cubes in all.**
(a) Write the missing number on the top view: ☐
(b) You can see 5 cubes in the picture. You cannot see ☐ cube.

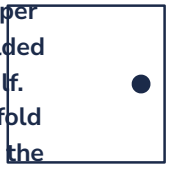
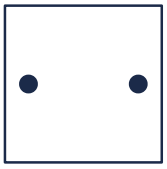
2	1
1	

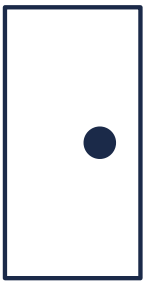


Answer: 2; 1

Hint (say this first): Add the numbers you can read. Take that total away from 6.
Worked solution: The top view has 2, 1 and 1. 2 + 1 + 1 = 4. The stack has 6 cubes, so the missing number is 2. The picture shows 5 cubes, so 1 cube is hidden.
Common mistakes: Writing 6 for the missing number.; Saying no cube is hidden.
Time for a middle child: about 5 min.

T2 **A paper is folded in half. The fold is on the left. One hole is punched.**



(a) There are ☐ holes when it is opened.
(b) Circle the opened paper: first, second or third.

Answer: 2; second

Hint (say this first): The paper has two layers. The holes are the same distance from the fold.
Worked solution: The fold is in the middle of the opened paper. One punch goes through 2 layers, so there are 2 holes, one each side of the fold and the same distance from it. That is the second picture. The first has holes on the same side. The third has only 1 hole.
Common mistakes: Writing 1 hole.; Choosing two holes on the same side of the fold.
Time for a middle child: about 5 min.

T3 **A paper was folded along the dashed line and one hole was punched. The picture shows the opened paper. One hole is ringed.**



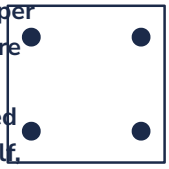
(a) Draw the other hole.
(b) The other hole is ☐ step from the dashed line.

Answer: 1 step on the other side of the dashed line

Hint (say this first): Count the steps from the ringed hole to the dashed line.
Worked solution: The ringed hole is 1 step to the left of the dashed line. The other hole is 1 step to the right of it, on the same row. Draw a hole on that dot.
Common mistakes: Drawing the hole on the same side.; Counting dots instead of steps.
Time for a middle child: about 4 min.

Go further (page 4): mastery and extension

G1 **A paper square is folded in half, then in half again. One hole is punched through all the layers.**



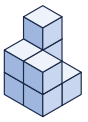



(a) There are ☐ holes when it is opened.
(b) Circle the opened paper: first, second or third.

Answer: 4; third

Hint (say this first): Each fold doubles the layers. 1 layer, then 2, then 4.
Worked solution: After the first fold the paper has 2 layers. After the second fold it has 4 layers. One punch goes through 4 layers, so there are 4 holes, one in each part of the opened paper. That is the third picture.
Common mistakes: Writing 2 holes.; Choosing 4 holes in a row.
Time for a middle child: about 6 min.

G2 **A stack has a 2 by 2 base. The back stacks have 3 and 1 cubes. The front left stack has 2. The stack has 8 cubes in all.**



(a) The front right stack has ☐ cubes.
(b) You cannot see ☐ cube in the picture.

Answer: 2; 1

Hint (say this first): Add 3, 1 and 2. Take the total away from 8.
Worked solution: 3 + 1 + 2 = 6. 8 take away 6 is 2, so the front right stack has 2 cubes. The picture shows 7 cubes, so 1 cube is hidden behind the front right tower.
Common mistakes: Writing 8 for the front right stack.; Saying no cube is hidden.
Time for a middle child: about 6 min.

G3 **Build a stack with 7 cubes on a 2 by 2 base. No stack is taller than 3 cubes. Many answers are right.**



(a) Write how many cubes are in each stack on the top view.
(b) The four numbers make ☐ cubes in all.

Answer: Many answers are right: four numbers from 1 to 3 that add to 7, for example 3, 2, 1, 1

Hint (say this first): Try 3 for one stack. Then find three more numbers that make 4.
Worked solution: For example 3, 2, 1 and 1: 3 + 2 + 1 + 1 = 7. Another answer is 2, 2, 2 and 1. Every stack has at least 1 cube.
Common mistakes: Four numbers that do not add to 7.; A stack taller than 3.
Time for a middle child: about 5 min.

Investigation (page 5): Fold, mark and open

Easy first step: Fold a square in half. Mark one dot away from the fold. How many dots when opened? ____
Then: Open it. Write one thing that is the same about the dots.

Model answer: There will be 2 dots. The two dots are the same distance from the fold line, one on each side. The paper had two layers, so one dot made two.

Look for: A child who says the dots are the same distance from the fold, or are opposite each other across the fold line. Credit any clear reason. A child who folds again and finds 4 dots has gone further.
Common mistakes: Saying 1 dot.; Putting the dot on the fold line.

Talk and explain (minutes 70 to 76)

A child in another class counts the cubes she can see in a picture. She sees 6 cubes. She says there are 6 cubes in the stack. Do you agree? How could we check? Sentence starter: I think ... because ...

Model answer: No. A picture can hide a cube behind another cube, so counting only what we see can give too few. We can check with blocks, or with a top view and add the numbers.

Look for: A child who says a cube can be hidden, or who says to build it or to use the top view. Not just yes or no. Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

Exit ticket (minutes 76 to 82): 3 questions

E1 **Look at the cubes.**



There are ☐ cubes in the picture.

Answer: 4 cubes

Worked solution: Touch each cube once: 4 cubes. No cube is hidden in this picture.
Common mistakes: Counting a cube twice.; Counting faces instead of cubes.

E2 **The top view shows how many cubes are in each stack.**



There are ☐ cubes in all.

Answer: 6 cubes

Worked solution: 2 + 1 + 2 + 1 = 6 cubes.
Common mistakes: Counting the squares (4).; Adding only the top row.

E3 **A square is folded so the left edge meets the right edge.**



Draw the fold line on the square.

Answer: A line down the middle, from the top edge to the bottom edge

Worked solution: When the left edge meets the right edge, the paper bends halfway across. The fold line goes down the middle of the square, from the top edge to the bottom edge.
Common mistakes: Drawing the line across the middle.; Drawing the line on one side.

Look-back (minutes 82 to 85)

Children tick or write: *Something I did well*; *The hardest part was* (hidden cubes; top views; folding; something else); *What helped me* (my blocks; top view numbers; folding paper; I was not stuck); *Next time I will try* (hidden cubes; top views; holes; asking a question); *I am still stuck (my grown-up will photograph this page for my tutor)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

5. Home practice and online practice

#	Tier	Question	Answer
H1	Start here	The top view of a stack shows 2, 1, 3 and 2 cubes. How many cubes are in the stack? There are ___ cubes.	8
H2	Start here	A stack is 3 cubes tall. How many cubes are under the top cube? There are ___ cubes.	2
H3	Start here	Lily has 12 blocks. She builds a stack with 7 blocks. How many blocks are left? There are ___ blocks left.	5
H4	Think harder	A paper is folded in half and one hole is punched. When it is opened, one hole is 3 steps from the fold line. How many steps from the fold line is the other hole? It is ___ steps.	3
H5	Start here	Ali has 13 counters. He puts them in pairs. How many counters are left over? ___ counter is left over.	1

Online practice (`out/l34/practice`, key `t4l34`): 13 items; 7 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no scores, no streaks.

6. Parent line for the term report

Your child is learning to picture things from the top and when folded. This week: counting every cube in a stack, reading a top view, and how many holes a folded paper makes.
Term line (plan, section 5): Your child is learning to choose a way to solve a problem, and to make and test their own puzzles. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

7. Slide list

#	Time	Slide	On screen	Speaker note
1	0–2	**Cubes and Views** s-title	TODAY WE WILL: / Count every cube, even one we cannot see. / Look at a stack from the top. / Fold a paper and find the holes.	Lesson 4 of Term 4, 90 minutes. Objective: count all the cubes in a stack including the one you cannot see, read a top view as a grid of numbers, and say how many holes a folded paper has when it is opened. Plain statement for families: counting cubes in a picture is early work; 3-D shapes are Primary 2 and drawing cubes is Primary 5, so reading a top view as numbers, the cube you cannot see and fold and punch go beyond the Primary 1 syllabus on purpose, always with real blocks and paper in hand. Time plan: opening 0 to 2, warm-up 2 to 8, sneak-peek 8 to 12, explore 12 to 23, mini-lesson 23 to 34, brain break 34 to 38, workbook 38 to 70, talk 70 to 76, exit ticket 76 to 82, look-back 82 to 85, flex time and goodbye 85 to 90 (the buffer: do not start anything new). Think harder, Go further and the Investigation are for every child to choose. Say the recording notice before you start the recording, in one line: you, the shared screen and any unmuted voice; a child may keep the camera off.
2	2–8	**Warm-up: rows** s-rem1	[picture: array] / Count the counters by rows. How many rows? How many in each row? How many in all?	Counting by rows (Term 2 Lesson 5, taught again). 2 rows with 3 counters in each row: 3 and 3 make 6. Ask: how many rows (2), how many in each row (3), how many in all (6). A child who is unsure points along each row. Your turn, no picture: 3 rows of 2; how many in all (6). Children answer aloud. No writing.
3	2–8	**Doubles** s-rem2	[picture: objects] / A double is the same number twice. Double 3 is 3 and 3 again. Double 5?	Doubles (Term 2 Lesson 3, taught again). Double 3 is 3 and 3 again: 6. Ask: what is double 3 (6), what is double 5 (10). A child who is unsure holds up 5 fingers on each hand. Your turn: double 4 (8). Say: each time we double, the number of layers doubles; we use that later. Children answer aloud. No writing.
4	2–8	**Count the cubes** s-rem3	[picture: cube_stack] / Touch each cube. How many cubes are there?	Counting cubes in a picture (new picture, taught with a finger). The picture shows a tower of 2 cubes with 1 cube next to it: 3 cubes in all. Ask: how many cubes (3). A child who is unsure touches each cube once. Your turn: the same shape with one more cube on the small tower (4). Children answer aloud. No writing.
5	8–12	**Sneak-peek puzzles** s-sneak	Ideas we have not learnt yet. A guess is a great start. Nobody is marked. / PUZZLE 1: anyone can try: / How many cubes are in the picture? / Write the number. / [picture: cube_stack] / PUZZLE 2: think: / The stack has 5 cubes. / The picture shows only 4. / Where is the 5th cube? I know because ___. / [picture: cube_stack]	Puzzle 1 is an access question: 2 cubes side by side and 2 on top, a wall of 4 cubes: the answer is 4. Puzzle 2 is a thinking question on the lesson idea: the stack has 5 cubes but the picture shows only 4, because one cube is hidden behind the front tower. A child who says 'it is behind the tall tower' has done well. Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write on their own paper. Nobody holds paper to the camera. There is no private chat. Answers: 4; the 5th cube is hidden behind the front tower. Do not reveal puzzle 2 until Talk time. Do not write scores or keep a list.
6	12–23	**Explore: build the stack** s-explore1	[picture: cube_stack] / [picture: cube_stack] / The numbers say how many blocks are in each stack. Build the stack. How many blocks do you use?	Round 1, about 5 minutes. Each child has up to 12 small blocks of one size (bricks or dice). The slide shows a stack on a 2 by 2 base and its top view: the numbers say how many blocks are in each stack, the back stacks in the top row. Children build it and count their blocks. Ask: how many blocks did you use; how many blocks can you see in the picture; how many can you not see. The back left bottom block is hidden behind the front right tower: the picture shows 5 blocks and the stack has 6. Answer on the next slide. A child with no blocks puts counters or buttons on a drawn 2 by 2 grid: the number of buttons in a square is the height of that stack. Nobody holds blocks or drawings up to the camera: children say how many they used and show fingers.

7	12–23	**6 blocks, 1 hidden** s-explore1-ans	[picture: cube_stack] / The stack has 6 blocks. The picture shows only 5. One block is hidden behind the front tower.	Children check their own stack. The stack has 6 blocks: 1 + 1 + 2 + 2. The picture shows only 5; one block is hidden behind the front right tower. Say: a picture cannot show every block, but the top view numbers let us count them all. Ticks only. Do not name any child.
8	12–23	**Explore: fold and open** s-explore2	Fold your paper square in half. Make the edges meet. / Make one pencil dot on the folded paper, away from the fold. / How many dots will there be when you open it?	Round 2, about 4 minutes. Each child has a paper square from the kit page (cut out by a grown-up beforehand). Read the three steps one at a time. Children fold the square in half so two edges meet, press the fold, and make one pencil dot on the folded paper, away from the fold edge. No punching: pencil only. Ask: how many dots will you see when you open it. Take guesses, then children open the paper. Nobody holds the paper up to the camera: children say what they see. A child with no square folds any sheet of paper.
9	12–23	**Open it up** s-explore2-ans	[picture: fold_punch] / One dot on the folded paper. Two dots when it is open.	The picture shows an opened paper: two dots, one each side of the fold. The dots are the same distance from the fold line, one on the left and one on the right. Say: the paper had two layers, so the dot went through two layers. Children check their own paper: two dots, one each side of the fold line, the same distance from it. Ticks only.
10	12–23	**What did we notice?** s-explore3	What was hidden when you looked at the picture of the stack? / Where were the two dots after you opened the paper?	Take one or two answers from volunteers. Key ideas: some blocks are hidden in a picture but they are still there; we can count them from the top view; the folded paper had two layers, so one dot made two dots, one on each side of the fold and the same distance from it. Repeat each child answer in your own words so the recording picks it up. Do not give the rule before the children say it; the mini-lesson makes it exact.
11	23–34	**Count every cube** s-m1	[picture: cube_stack] / [picture: cube_stack] / The top view says how many cubes are in each stack. How many cubes can you see in the picture? How many are there in all?	Worked example. The slide shows a stack on a 2 by 2 base and its top view. The top view numbers add to 7 cubes; one cube is hidden behind the front right tower, so 6 can be seen in the picture. Ask: how many cubes can you see (6); how many cubes are there in all (7). The answer is on the next slide. Say: a cube we cannot see is still a cube. Children may build it with blocks to check.
12	23–34	**7 cubes in all** s-m1-ans	[picture: cube_stack] / We can see 6 cubes. There are 7 in all: 2 + 2 + 1 + 2. One cube is hidden.	The answer for the example on the slide before. 6 cubes can be seen, 7 in all. The top view shows 2, 2, 1 and 2: the four stacks. Read it aloud so that children can tick their own work, and say what the picture showed. Your-turn answer: not needed here, the next slide uses the top view.
13	23–34	**The top view as numbers** s-m2	[picture: cube_stack] / Look from the top. Write a number for each stack. Draw 4 squares like a window.	Worked example. A stack with 6 cubes and none hidden: the back left stack has 1, the back right 1, the front left 3 and the front right 1. Children draw four squares in a 2 by 2 block on paper and write how many cubes are in each stack, the back stacks in the top row. The answer is on the next slide. Say: the top view is a plan of the stack, seen from above; the numbers say how tall each stack is. A child who writes the cubes they can see on each side has counted faces: ask, how many cubes are in this stack?
14	23–34	**The top view** s-m2-ans	[picture: cube_stack] / [picture: cube_stack] / Top row: 1 and 1. Bottom row: 3 and 1. There are 6 cubes in all.	The answer for the example on the slide before. The top row is the back of the stack and the bottom row is the front. 1, 1, 3 and 1 make 6 cubes. Read it aloud so that children can tick their own work.
15	23–34	**Build from the top view** s-m3	[picture: cube_stack] / Build the stack with your blocks. How many cubes are there in all?	Worked example. The top view says: the back left stack has 3 cubes, the other three stacks have 1 cube each. Children build it with blocks, or put buttons on a drawn 2 by 2 grid, and count: 3 and 1 and 1 and 1 make 6 cubes. The answer is on the next slide. Say: a top view tells us how many cubes without seeing the side. If a child builds a stack of 3 and forgets the others, ask: what does each number say?

16	23–34	**6 cubes** s-m3-ans	[picture: cube_stack] / 3 + 1 + 1 + 1 = 6 cubes. The tall stack is at the back on the left.	The answer for the example on the slide before. 3 + 1 + 1 + 1 = 6. The picture shows the stack seen from the front: the tall stack is at the back on the left, and its bottom cube is hidden behind the front cubes. Read it aloud so that children can tick their own work.
17	23–34	**Fold once, punch once** s-m4	[picture: fold_punch] / A paper is folded in half. The fold is on the left. One hole is punched. How many holes when it is opened?	Worked example. The picture shows the folded paper with one hole punched through both layers. The folded edge is on the left. Ask: how many holes will there be when it is opened (2). Where will they be (one on each side of the middle line, the same distance from it). The answer is on the next slide. Children use their paper square: fold, make one pencil dot, open it, check. No real punching.
18	23–34	**Two holes** s-m4-ans	[picture: fold_punch] / 2 holes. The fold is in the middle. The holes are the same distance from it.	The answer for the example on the slide before. The opened paper has 2 holes, one each side of the fold line and the same distance from it. Say: the paper had two layers, so one punch made two holes. Read it aloud so that children can tick their own work.
19	23–34	**Same distance from the fold** s-m5	[picture: grid] / The paper is folded along the dashed line. One hole is ringed. Where is the other hole?	Worked example. A dot grid stands for the opened paper, and the dashed line is the fold. The ringed hole is 2 steps to the left of the dashed line. Ask: how many steps from the dashed line is the ringed hole (2); where is the other hole (2 steps to the right of the dashed line, on the same row). The answer is on the next slide. Link to the grid lesson: count the steps, not the dots.
20	23–34	**2 steps each side** s-m5-ans	[picture: grid] / The other hole is 2 steps from the dashed line, on the other side.	The answer for the example on the slide before. Both holes are 2 steps from the dashed line, on the same row. Read it aloud so that children can tick their own work.
21	34–38	**Fold and unfold break** s-break	Stand up. Fold your arms in half. Unfold them. / Make a tall stack with your arms. Make a flat one. / Sip some water.	Four minutes. Stand up on camera with a little clear space. Call movements: fold your arms in half, open them, stand tall like a stack of 3, crouch like a stack of 1, make a square with your arms. Then fold in half again and again, with the body bending, and say how many layers: 2, then 4 (each fold doubles the layers). No equipment, no running. A child may stay seated. Do not give prizes or compare children.
22	38–70	**Workbook page 1: Start here** s-ws1	Questions 1 to 3. Take your time. / questions S1, S2, S3	Read questions 1 to 3 aloud. Children work on their own; the tutor walks the cameras and asks how do you know. Wait 10 seconds before helping. After question 3 any child may choose to carry on or try Think harder. Question 1: count the cubes in the picture. Question 2: how many cubes can you not see. Question 3: which top view matches the stack.
23	38–70	**Check page 1** s-ws1ans	answers S1, S2, S3	Read each answer aloud. Children tick their own page with a tick. Never ask a child to show a wrong answer to the group. Answers: Q1: 5 cubes. Q2: 4 cubes in all, 1 you cannot see. Q3: B.
24	38–70	**Workbook page 2: Start here** s-ws2	Questions 4 to 6. Take your time. / questions S4, S5, S6	Read questions 4 to 6 aloud. Question 4: write the number of cubes in each stack on the top view. Question 5: a paper square is folded so the top edge meets the bottom edge; draw the fold line. Question 6: a top view with numbers; how many cubes in all. Wait 10 seconds before helping.
25	38–70	**Check page 2** s-ws2ans	answers S4, S5, S6	Read each answer aloud so a child watching the recording can tick their own work. Never ask a child to show a wrong answer to the group. Answers: Q4 1, 3 in the top row and 1, 1 in the bottom row. Q5 a line across the middle, from the left edge to the right edge. Q6 7 cubes.

26	38–70	**Choose your next page** s-choice	Page | What it is / Think harder (page 3) | Hidden cubes, a fold and a hole, the other hole / Go further (page 4) | Fold twice, a bigger stack, build your own / Investigation (page 5) | Fold, mark and open / Check | Finish and check your own work / Everyone may choose. Not finished? Pages 3 to 5 are for home too.	Invite the whole group at once, for example: when you have finished Start here, you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. Pairs are not split unless both children agree. Model answers are in the tutor guide. About 5 minutes.
27	70–76	**Talk and explain** s-talk	A CHILD IN ANOTHER CLASS SAYS: / I count the cubes I can see in the picture. / I see 6 cubes. / So there are 6 cubes in the stack. / Do you agree? How could we check? / I think … because …	One minute of talk to the grown-up next to you, or to your screen. Volunteers first; a pass is always fine; never call a child by name to answer. Model answer: no, we do not agree. A picture can hide a cube behind another cube, so counting only what we see can give too few. We can check with blocks, or with a top view: add the numbers. Link to the sneak-peek and the mini-lesson. Repeat each child answer in your own words for the recording. The child in the story is in another class: no cast name is given the wrong answer.
28	70–76	**Back to the sneak-peek** s-sneakans	Puzzle 1: 4 cubes. Puzzle 2: the 5th cube is hidden behind the front tower.	Reveal both puzzles now. Say: many of you wrote a good answer. Do not read out anyone earlier answer. Puzzle 1: a wall of 4 cubes, all can be seen. Puzzle 2: the picture shows 4 cubes, but there are 5: one is hidden behind the front tower, at the back. It was the same idea as Explore and the mini-lesson: a cube we cannot see is still a cube.
29	76–82	**Exit ticket** s-exit	On your own. No looking back. / questions E1, E2, E3	Children answer on page 5. Read E1, E2 and E3 aloud, with the numbers (the figures on the slide are small). Children tick their own ticket against the next slide. Nobody holds paper to the camera and there is no private chat.
30	76–82	**Check the exit ticket** s-exitans	answers E1, E2, E3	Read each answer aloud. Children tick their own ticket with a tick. Answers: E1 4 cubes; E2 6 cubes; E3 a line down the middle, from the top edge to the bottom edge.
31	82–85	**Look-back** s-look	[look-back box]	Three minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never marked, never collected in front of the group and never read out. If a child ticks I am still stuck, a grown-up may photograph that page after class and send it to the business WhatsApp (the number is in the parent summary, never on a slide). You then help in the open session next lesson. Never a direct message to a child. The Next time I will try row is optional.
32	85–90	**Well done, inquirers!** s-end	Show a grown-up a stack and its top view. / NEXT LESSON: / Spot It / Have ready: the 12 pair cards from your book, cut out by a grown-up.	Say goodbye and thank the grown-ups. Minutes 85 to 90 are flex time: the buffer. If you ran late, finish any open item (the look-back). Do not start anything new. Remind families that the next lesson needs the pair-cards page of Book 7, torn out along the dashed line, with the 12 cards cut out by a grown-up beforehand. If the whole class is ahead, show the optional slides and use the Spare-time menu from the tutor guide.
33	optional	**Go further: fold twice** s-opt	[picture: fold_punch] / Folded in half, then in half again. One hole. How many holes when it is opened?	Optional, only if you are on time. A paper is folded in half, then in half again, so there are 4 layers; one hole is punched through all of them. Ask: how many holes when it is opened (4). Each fold doubles the layers: 1, 2, 4. Take two answers and move on. This links to doubles in the warm-up.

34	optional	**If we have time: pick one** s-menu	Activity | What we do / Is it always true? | Does a bigger stack always have more cubes in the top view numbers? / Make a puzzle | Build a stack and draw its top view for your grown-up. / Teach it back | Show a toy why a picture can hide a cube. / Optional. Everyone may join.	Optional Spare-time menu, used only if the whole class is ahead. Pick one or two activities (3 to 5 minutes each); full tutor lines and answers are in the tutor guide. Never start new content and never name who is ahead. Slow down with how do you know rather than rushing on. These minutes come from the flex buffer and are not part of the 90-minute sum.

8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

Chapter	Part	You say (recording voice)	A child watching later
0:00	Opening	*If you are watching the recording: hello! Get a pencil, paper, your workbook, your blocks and a paper square. Pause or rewind any time.*	Gather your workbook, blocks, paper square, pencil and paper. Press play when ready.
2:00	Remember warm-up	*Recording: join in! Say your answers out loud before I say them. Pause if you need longer.*	Play along. Say each answer out loud before the video does.
8:00	Sneak-peek puzzles	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*	Pause. Try both puzzles on your own paper. A guess is fine.
12:00	Explore	*Recording: pause now. Build the stack with your blocks, then fold your square and make one dot. Press play when you have tried.*	Pause. Build the stack and count. Fold a square, make one dot, then open it.
23:00	Mini-lesson	*Recording: keep watching and copy the examples with your blocks and paper. Rewind if you miss a step.*	Watch and copy the examples. Rewind any step you missed.
34:00	Brain break	*Recording: stand up and join in! Fold your arms in half, then open them.*	Stand up. Fold and open your arms, then sit down.
38:00	Workbook	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*	Pause. Do page 1, then play to check. Then page 2. Then choose a page to try.
70:00	Talk time	*Recording: pause. Tell a grown-up why counting only the cubes you can see can give too few.*	Pause. Explain your answer to someone at home and show how you know.
76:00	Exit ticket	*Recording: pause and do the exit ticket on your own. Then press play to check it.*	Pause. Do the exit ticket alone, then play to check it.
82:00	Look-back	*Recording: pause and fill in your look-back box. You can pass.*	Pause. Fill in the look-back box, or pass.
85:00	Flex time and goodbye	*Recording: well done for finishing! Show a grown-up a stack of blocks and its top view.*	Show a grown-up a stack and say its top view numbers.

9. Timing check (model, not a stopwatch)

Part	Allocated	Script	Slow class	Flag
opening	2:00	1:35	1:47	ok
remember	6:00	5:00	5:58	ok

sneak	4:00	3:00	3:31	ok
explore	11:00	9:52	11:43	OVER (slow class)
mini	11:00	10:10	12:01	OVER (slow class)
break	4:00	3:16	3:53	ok
work	32:00	23:55	28:30	ok
talk	6:00	4:26	5:16	ok
exit	6:00	4:23	5:10	ok
look	3:00	2:06	2:27	ok
flex	5:00	0:24	0:27	ok
Total	90:00	68:06	80:43	

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 6 questions; a middle child needs about 8.5 min for page 1 and 8.5 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.